

Real-Time Analytics Pipeline: Engineer 004

Brief version: 2026-07 | Henry Hai Nguyen

Fixture worked from: `fixtures/event_sample.jsonl`, SHA-256 **[Observed]**

`1aeb24b415009e89fcf8acb5a178410faf216dc17b16920d9849ecc8bbb24235`

Artifact: <https://github.com/henry-hai/beat-claude-engineer-004>

4 pages of prose; diagrams excluded from the count per the brief.

Written answer

Every number here regenerates from `pipeline/run.py`, `pipeline/latency.py` and `pipeline/cost.py`; outputs are committed in `results/*.json` **[Observed]**.

Thesis: the risk is at ingest, not in the broker

Every answer to this brief will open by picking a broker. That is the least interesting decision on the table. Kinesis, MSK and self-managed Kafka all move bytes, all publish pricing, all carry an SLA. Any of the three would work here, which is exactly why it is not where the risk lives. I still pick one below, and the deciding reason turns out to be operational rather than technical.

What will hurt you is in the ingest path, and the sample data proves it. Three events break a constraint the brief calls non-negotiable, and none look broken at a glance **[Observed]**, all traceable in `results/findings.json`:

- **evt-0009** uses an older SDK's field names (`timestamp`, `page_path`, `ref`, `type` `pageview`). Customers cannot be made to upgrade, so that shape is permanent. Reject it and a paying customer silently loses data.
- **evt-0007** carries a raw email and phone while `user_id` is null; **evt-0022** resolves that visitor to `u-7304` fifteen events later. And **evt-0017** demands GDPR deletion for `u-1077`, whose history includes **evt-0006**, written while anonymous. Delete by `user_id` and you miss it; an incomplete deletion is a regulatory failure, not a bug.
- **evt-0011** has no `tenant_id` **[Observed]**. Across 500 tenants, guessing is not a data quality issue; it is one customer seeing another's visitors.

So ingest is the component I built, ran, and measured.

1. Architecture and technology choices

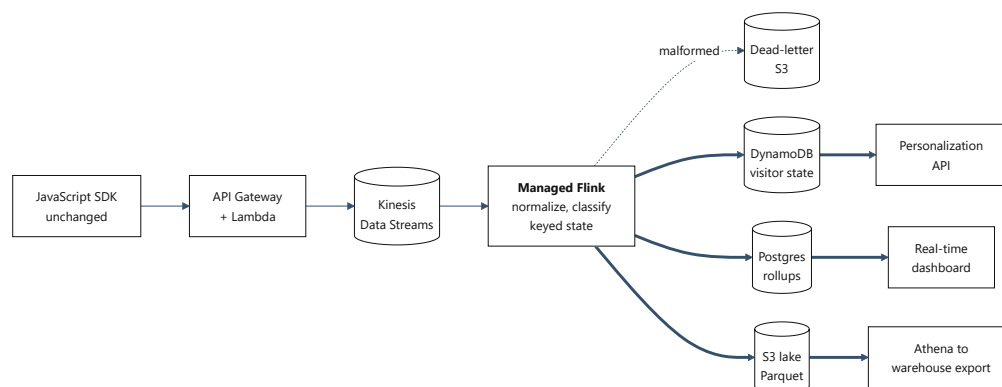


Figure 1. SDK to dashboard. The dead-letter branch is drawn deliberately: a malformed record must never stop the consumer.

Why each component, and what was rejected

COMPONENT	CHOSEN	REJECTED	DECIDING REASON
Ingest	Kinesis Data Streams	MSK, self-managed Kafka	Modeled infrastructure is 3.6% of the ceiling [Estimated] , so engineer-hours are scarcer than dollars. MSK is cheaper per byte and costs broker operations two engineers do not have.
Processing	Managed Flink	Lambda, ECS consumers	Four of the thirteen checks the detector performs cannot be answered from a single event [Observed] . Those need checkpointed keyed state, which a stateless consumer cannot provide at any price.
Visitor state	DynamoDB	Redis-only, Postgres	Single-digit-millisecond point lookups on the personalization path, per-tenant partition keys, no capacity planning.
Dashboard serving	Postgres rollups	ClickHouse, OpenSearch	The team already runs PostgreSQL. Aggregation happens upstream, so it stores summaries rather than raw events. Reversible; see the tripwire in section 3.
Event lake	S3 + Athena	Redshift	The brief requires export to Snowflake and BigQuery; Parquet on S3 is the neutral format both read.
Hot cache	ElastiCache Redis	none	Already in their stack. Segment membership, not raw events.

Unit prices and per-component lines: `results/cost_model.json` [\[Benchmarked\]](#).

The **stateless/stateful split is the load-bearing choice**. Normalization and per-event classification are pure functions of one event, so any worker can process any record. Duplicate detection, burst detection, identity stitching and deletion scope are not; they depend on what that worker has seen. That boundary, not the broker, decides whether this needs Flink-class state: 9 checks on one side, 4 on the other [\[Observed\]](#).

Event structure and identity

One canonical envelope (`event_id`, `tenant_id`, `anonymous_id`, `user_id`, `type`, `ts`, `received_at`, `properties`), with legacy names mapped at the edge from a lookup table rather than in code, so registering the next SDK generation is a config change. Every transformation is recorded, so the pipeline can always answer *what did you change about this event?* `ts` is the browser's clock, `received_at` is ours; both are kept and neither "corrects" the other.

Identity resolves backwards. A visitor browses anonymously, signs in, and their earlier events become theirs retroactively. Four such stitches appear in 25 lines [\[Observed\]](#), including `anon-3d0` to `u-7304`, which retroactively attaches `evt-0007`'s personal data to a named person. Deletion therefore runs against the identity graph in durable storage, never the bounded in-memory window, which returns a lower bound rather than the truth. See `pipeline/state.py`.

2. Scale, reliability and migration

Sizing, and the number that changes the answer

50M events/day [\[Observed\]](#) is 579 events/second average, 5,787 at the stated 10x spike [\[Estimated\]](#). At the 229-byte mean event size measured on the fixture [\[Observed\]](#), that peak is 1.33 MB/s.

A Kinesis shard gives 1 MB/s or 1,000 records/second [\[Benchmarked\]](#). The peak therefore needs 1.3 shards by bandwidth and 5.8 by record count [\[Estimated\]](#). **Record count binds.** A shard saturates its record limit while using 23% of its bandwidth, so sizing on throughput under-provisions by roughly 4x. That is one plausible mechanism behind "crashes during traffic spikes." Provisioned at 8 shards, 30% headroom [\[Assumed\]](#); derivation in `results/cost_model.json`. That result exists only because the event size was measured, not guessed.

Zero loss, and what degrades first

Delivery is at-least-once, so dedupe keys on `(tenant_id, event_id)`, not `event_id` alone, because SDKs do not coordinate IDs across customers and collapsing them deletes one tenant's event on another's collision. `evt-0002` appears twice, 7.4 seconds apart: a retry [\[Observed\]](#).

Unparseable records go to a dead-letter sink with bytes intact, never the floor. `evt-0020` is one missing brace; the obvious consumer dies there and never reads the four events after it [\[Observed\]](#). Scale that to a crash-looping consumer while the broker keeps accepting writes and you have the loss mechanism the brief describes. Under overload the system sheds in a stated order and **ingest never sheds**: rollup granularity coarsens, then personalization falls back to cached segments, then lake writes batch larger. The broker is the buffer; buffered data is late, not lost. See `pipeline/normalize.py`.

Migration: per-tenant, with an automatic trigger

The SDK cannot change and 500+ customers cannot break [\[Observed\]](#), so no part of the cutover is visible to them. API Gateway forks to both pipelines with the old one authoritative; both process everything in parallel; shadow dashboards go internal-only; then per-tenant cutover, smallest first. "Accuracy verified" means, per tenant-hour: counts agree within 0.1% [\[Assumed\]](#) and an `event_id` join leaves no unmatched rows on either side.

The rollback trigger is automatic and needs no human. Any tenant-hour diverging more than 0.5% for two consecutive hours [\[Assumed\]](#), or dashboard p95 over 5 seconds, reverts that tenant and pages on-call. Rollback is per-tenant, so a bad cutover costs one customer an hour rather than all of them. The comparison runs the same code as `pipeline/rules.py` against both streams.

Two accuracy signals run continuously afterwards [\[Assumed\]](#): the gap between `evt-0019`'s client-side counter and the server's own record, which is a free measurement of pipeline loss; and new instrumentation, a monotonic per-session sequence number, because gaps in a server-assigned `event_id` cannot detect loss at all. See section 5.

3. Trade-offs and risks

Optimizing for correctness you can audit and a system two engineers can operate; **sacrificing** peak cost efficiency and, in the MVP, query flexibility. Modeled infrastructure is \$1,809/month, 3.6% of the ceiling [\[Estimated\]](#) from [\[Benchmarked\]](#) AWS prices, and that survives every assumption being wrong by 28x. **The \$50K ceiling is not the binding constraint. Two engineers and three months are.** That inverts the usual build-versus-buy call: self-managed Kafka is cheaper per byte and spends the scarcer resource.

The MVP excludes a columnar serving tier, ML segmentation, per-tenant retention, real-time backfill, and cross-tenant benchmarking.

The serving tier is reversible, with a computed reversal condition. Dashboards come from pre-aggregated rollups in PostgreSQL, which this team already runs. At the modeled shape that table holds 100M hot rows against an assumed 250M ceiling: 40% utilization, 2.5x headroom [\[Estimated\]](#). Three conditions trigger a migration review: rows past the ceiling (leading), dashboard p95 over 1s across 7 days, or rollup lag over 30s. An option held, not a migration scheduled. It may never fire.

With more time I would measure the per-tenant distribution first. 50M/day is a fleet-wide total [\[Observed\]](#); 500 tenants at 100k is a different system from 20 at 2M and 480 at 8k [\[Assumed\]](#), and which it is changes the serving tier. Figures derive from `pipeline/cost.py`; the tripwire is in `results/cost_model.json`.

4. What is actually in the sample data

13 distinct anomaly classes across 25 lines: 9 answerable from a single event, 4 requiring memory of others [\[Observed\]](#). Full output with every `event_id` in `results/findings.json`.

CLASS	EVENTS	PIPELINE RESPONSE
Malformed JSON	evt-0020	Dead-lettered, bytes intact; consumer continues
Duplicate delivery	evt-0002	Dedupe on (tenant_id, event_id) before counting
Schema drift / no server clock	evt-0009	Normalized at the edge; received_at left null, never invented
Null tenant	evt-0011	Quarantined; never attributed by inference
Clock skew, 3 magnitudes	evt-0005, -0006, -0016	Ladder: 47s, 65min, 365d. Excluded from windowed math, not corrected
Far-future timestamp	evt-0016	Quarantined pre-windowing; would advance the watermark past every real window
Bot traffic	evt-0012 - -0015	Two signals: referrer, and 4 events in 51ms. Quarantined, never deleted
PII in properties	evt-0007	Caught by key name and value shape; tagged for deletion tooling
Privacy request in-stream	evt-0017	Routed to the deletion workflow, not counted as behavior
Deletion scope	evt-0017	Resolved via identity graph; reaches evt-0006, written anonymously
Identity stitching	evt-0003, -0008, -0017, -0022	History re-attributed retroactively
Client-asserted aggregate	evt-0019	Stored with provenance, never source of truth. See section 5
all rows	25 lines	[Observed], regenerate via pipeline/run.py

The 9/4 split is itself an architectural finding. Those four cannot be answered by a stateless consumer at any price. They are the argument for keyed state, and why the detector separates them rather than hiding the boundary.

5. Three things I would not act on at face value

All three are reproducible against results/findings.json.

1. The number the browser made up. The brief asks for a "viewed pricing 3x" segment. evt-0019 supplies exactly that: count_today: 3. The server's own record for that visitor, anon-9f2, shows one pricing page view [Observed]. A counter computed in a browser cannot be audited, cannot be recomputed, and is trivially forged. The trap is not spotting bad data. It is that the brief asks for the feature the bad data would break. Segments are recomputed server-side. The claim is still stored, in a separate namespace, because the divergence is a free loss measurement.

2. The bot cluster. evt-0012 - evt-0015: four page views in 51ms from one visitor, all referred by a host containing "scanner" [Observed]. I flag and quarantine; I do not delete. A referrer is attacker-controlled and legitimate hosts contain that substring. Wrong and deleted, real customer data is gone with no recovery; wrong and labeled, the cost is one column in a table.

3. The brief's own loss figure. It states ~3% loss at peak. This fixture can neither confirm nor refute that, and I will not pretend otherwise. event_id runs evt-0001 to evt-0024 with no gaps [Observed], but a curated sample shows no gaps whether or not the system leaks, and we do not know whether event_id is assigned client-side or on receipt. If the server assigns it, loss is invisible by construction. **What would settle it:** a monotonic per-session sequence number, emitted by the SDK and reconciled server-side. That is why it is in week-one scope rather than a "we would monitor this" sentence.

Required Submission Packet. The written answer ends here. The sections below are the packet items the brief lists separately: operating artifact, artifact access, evidence log, number source labels, AI usage disclosure, failure modes, and what stays human.

Operating artifact

A runnable ingest-validation pipeline in pure Python 3, standard library only. No dependencies, no Docker, no network. It is the prototype of the component argued above to be the riskiest, and it doubles as the ingest stage of the design rather than being a separate demo.

Four modules, split along how stream processors actually execute work: `normalize.py` (stateless map), `rules.py` (stateless classify), `state.py` (keyed aggregate over a window), `run.py` (job graph and sinks). Everything in the first two is a pure function of one event and lifts into production unchanged; everything that could not satisfy that contract was pushed into `state.py` rather than smuggled in.

Every number in this document regenerates from these three commands:

```
python -m pipeline.run      fixtures/event_sample.jsonl
python -m pipeline.latency fixtures/event_sample.jsonl
python -m pipeline.cost
```

Outputs are committed to `results/`, so findings are readable without running anything. 50 tests **[Observed]**: `python -m unittest discover -s tests -t .`

Artifact access

<https://github.com/henry-hai/beat-claude-engineer-004>. Public, verified reachable unauthenticated. `git log` is the build history in order.

```
git clone https://github.com/henry-hai/beat-claude-engineer-004
cd beat-claude-engineer-004 && python -m pipeline.run fixtures/event_sample.jsonl
```

The fixture is vendored so the repo runs standalone, and `.gitattributes` pins `*.jsonl` with `-text` so Git never rewrites its line endings. Without it, cloning on Windows converts 25 line endings to CRLF and moves the file from 5,749 to 5,774 bytes: identical text to a human, a completely different SHA-256 **[Observed]**. GitHub serves 5,749 bytes, so `shasum -a 256` reproduces the hash at the top of this document on any platform.

Evidence log

Tiers per SCORING.md. Every figure is **[Observed]** unless the row says otherwise.

CLAIM	TIER	CHECK IT
13 anomaly classes, 21 findings, every <code>event_id</code> cited	3	<code>results/findings.json</code>
Fixture checksum reproduces on any OS	3	<code>shasum -a 256 fixtures/event_sample.jsonl</code>
Mean event size 229 bytes, median 228, range 200-285	3	<code>results/cost_model.json</code>
Batch vs streaming staleness: 481.9s vs 0.164s median	3	<code>results/latency.json</code> , method in <code>pipeline/latency.py</code>
Per-event processing 0.017ms mean	3	<code>results/latency.json</code>
Record count binds Kinesis sizing, not bandwidth	3	<code>results/cost_model.json</code>
Modeled spend \$1,809/mo, 3.6% of ceiling, AWS prices dated	3	<code>results/cost_model.json</code>
Serving tripwire: 100M hot rows vs 250M ceiling	3	<code>results/cost_model.json</code>

CLAIM	TIER	CHECK IT
Build history and decision record	3	<code>git log</code> , 16 commits with reasoning
Architecture and rejected alternatives; 50-test suite	2	this document; <code>tests/</code>
Measured before/after from a comparable production system	none	I do not have this
<i>every figure above</i>	3	<code>[Observed]</code> via <code>results/*.json</code>

On the missing Tier 4. The rubric calls it the differentiator: measured before/after from a comparable system actually built or fixed. I have no production pipeline at this scale to cite, so I am not claiming it. The closest honest substitute is the latency harness in `pipeline/latency.py`, logged Tier 3 deliberately, because it is a benchmark run on a simulation I wrote, moving 25 events rather than 50 million `[Observed]`. Calling it Tier 4 would be inflation on a rubric built to catch exactly that.

Number source labels

Every number carries one of four labels, in the same paragraph. `[Observed]`: measured from the fixture or from running the artifact, reproducible with the commands above. `[Benchmarked]`: a published AWS list price, with page and retrieval date recorded in `pipeline/cost.py`. `[Estimated]`: derived by stated arithmetic from labeled inputs, with the derivation in the code rather than just the conclusion. `[Assumed]`: a placeholder chosen to make the plan concrete. Every threshold in the detector and every sizing assumption in the cost model is `[Assumed]` and lives as a named constant, easy to find and argue with rather than buried in a function.

AI usage disclosure

Tools. Claude Code (Opus 5) in a terminal for the build; a separate Claude browser session as an adversarial reviewer.

What AI did. Wrote the pipeline code, the cost model, and a first draft of this document from my direction. Pulled AWS list prices from published pages. Caught the line-ending problem that would have made my checksum unreproducible.

What I decided. That the fixture detector was the primary artifact and the broker POC optional. That the riskiest component is ingest, not transport. That my own prior work stays out of the evidence log, so every claim here is reproducible by a stranger. The voice. What the MVP excludes.

What I checked or changed. Every number was regenerated from the artifact, not transcribed. Two AWS components could not be priced from the published pages; rather than let a plausible figure through, I left them visibly unpriced. I used the second session to attack the first, and it correctly called out that I had claimed Tier 4 for the latency measurement when the rubric does not support it, hence Tier 3. Two model-generated numbers were wrong and caught by re-running the arithmetic; the rollup sizing used a cross-product where the real bound is traffic, which would have selected the wrong serving tier.

Known weak spots. The prose is fluent, and fluency is not evidence. Judge the repo, not the writing. Every threshold is `[Assumed]` rather than tuned against real data; they are named constants in `pipeline/rules.py` and `pipeline/state.py`, easy to find and argue with. I have not operated Kafka or Flink in production; this is reasoned from first principles and from the failure modes visible in the fixture, not from having run one at 50M events/day.

What breaks it

The sample is 25 events `[Observed]`. It cannot establish any rate: not error, not loss, not a distribution. Every conclusion from it is about mechanism, not magnitude.

Every threshold is assumed: clock-skew tolerances, the dedupe window, the burst threshold, the Postgres row ceiling. If real skew across the SDK fleet is distributed differently than guessed, the ladder mislabels events.

The bot heuristic has false positives by construction: a referrer is attacker-controlled and a legitimate company could be called anything. Which is why the pipeline labels rather than deletes.

Deletion scope from the artifact is a lower bound. Keyed state is bounded, so the in-memory graph knows only what it has recently seen. Real deletion runs against durable storage.

Two cost lines are unpriced, a managed OLAP tier and Aurora, because AWS did not render their rate tables when checked. With 96% of the ceiling unused [**Estimated**] they are unlikely to change the conclusion, but that is a prediction, not a calculation.

The per-tenant distribution is unknown, and it decides the serving tier. If the fleet is 20 large tenants rather than 500 even ones [**Assumed**], the Postgres rollup table is wrong from day one and the tripwire fires immediately rather than never.

The latency harness is a simulation. It has no network, no cluster, no queue. It isolates and measures one mechanism: batch-window size dominates end-to-end latency. It proves nothing about any broker. Method and exclusions are stated in `pipeline/latency.py`.

What stays human

Everything below is irreversible or legally binding. The detector labels rather than acts, precisely so these stay decisions.

Deleting suspected bot traffic. Automated flagging, human deletion. Wrong and deleted, there is no recovery; wrong and labeled, the cost is one column. Asymmetric risk means the automation stops at the flag.

The scope of a GDPR deletion. The pipeline can compute which anonymous history belongs to an identified user; it should not execute unreviewed. Both errors are unacceptable in different directions. Too little is a regulatory failure; too much destroys another person's data.

Overriding a rollback. The rollback is automatic and needs no human; that is the point. Pushing a tenant forward again after one fires is a judgment about whether the cause is understood.

Attributing a null-tenant event. `evt-0011` has no tenant [**Observed**]. Guessing which of 500+ customers owns it is a data breach if wrong, so the pipeline quarantines and a human resolves it against the source integration.

Declaring accuracy verified at cutover. The parallel run produces a number. Deciding it is good enough to move a paying customer is not a threshold check; it is accountability, and it needs a name attached.

Tuning any threshold here. Every one is [**Assumed**]. Changing them changes what the pipeline considers real, and that is never an autoscaling decision.